

NETFLIX

Content Insights

Exploratory Data Analysis

Understanding Netflix Movies & TV Shows through data

A business-focused EDA project

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Python • Pandas • NumPy • Matplotlib • Seaborn • Jupyter

Executive Summary

02

The story in one slide

THE QUESTION

What can Netflix's catalog tell us about its content mix, audience focus, geographic footprint and evolution over time?

THE EVIDENCE

The analysis looks at content type, ratings, content-added trends, release years, countries and directors using exploratory visualizations.

THE STORY

Netflix's catalog is movie-heavy, strongly concentrated in mature ratings and recent releases, with the US leading content volume and a sharp rise in additions around 2015–2019.

The purpose of EDA here is to turn catalog metadata into a clear business narrative.

Business Problem & Objective

03

Why this analysis matters

BUSINESS PROBLEM

Netflix has a large and diverse content library. Understanding how the catalog is distributed — and how it evolved — helps explain its content strategy.

BUSINESS OBJECTIVE

Analyze Netflix's Movies and TV Shows to identify important content patterns and trends that can support content-strategy discussions.

SUCCESS LOOKS LIKE

- A cleaned, analysis-ready dataset
- Clear exploratory visualizations
- Business-oriented interpretation of patterns
- Actionable recommendations grounded in the observed catalog

Dataset & Project Structure

04

What was analyzed

DATASET

Netflix Movies and TV Shows metadata. Key fields include title, type, director, cast, country, date added, release year and rating.

OUTPUT

The cleaned dataset is saved as `netflix_cleaned.csv`, while the full analysis is maintained in the Jupyter notebook: `Netflix EDA.ipynb`.

PROJECT PIPELINE

01

Raw data

02

Clean

03

Explore

04

Visualize

05

Interpret

Dataset source: Netflix Movies and TV Shows – Kaggle

Data Cleaning & Preparation

05

Making the raw metadata reliable for EDA

1

Missing values

Identify incomplete fields and handle them appropriately.

2

Data types

Fix data types so dates, numeric fields and categories can be analyzed.

3

Standardization

Standardize columns and values for consistent grouping and comparison.

4

Analysis-ready output

Save the processed dataset as `netflix_cleaned.csv`.

Why it matters: EDA conclusions are only useful when the underlying fields are consistent enough to compare.

Exploratory Data Analysis Workflow

06

From questions to evidence

01 UNDERSTAND

Inspect structure, fields and data quality

02 CLEAN

Prepare values and data types

03 UNIVARIATE

Study individual distributions

04 BIVARIATE

Compare categories and relationships

05 VISUALIZE

Use charts to reveal patterns

06 INTERPRET

Translate patterns into business insights

Business Questions

07

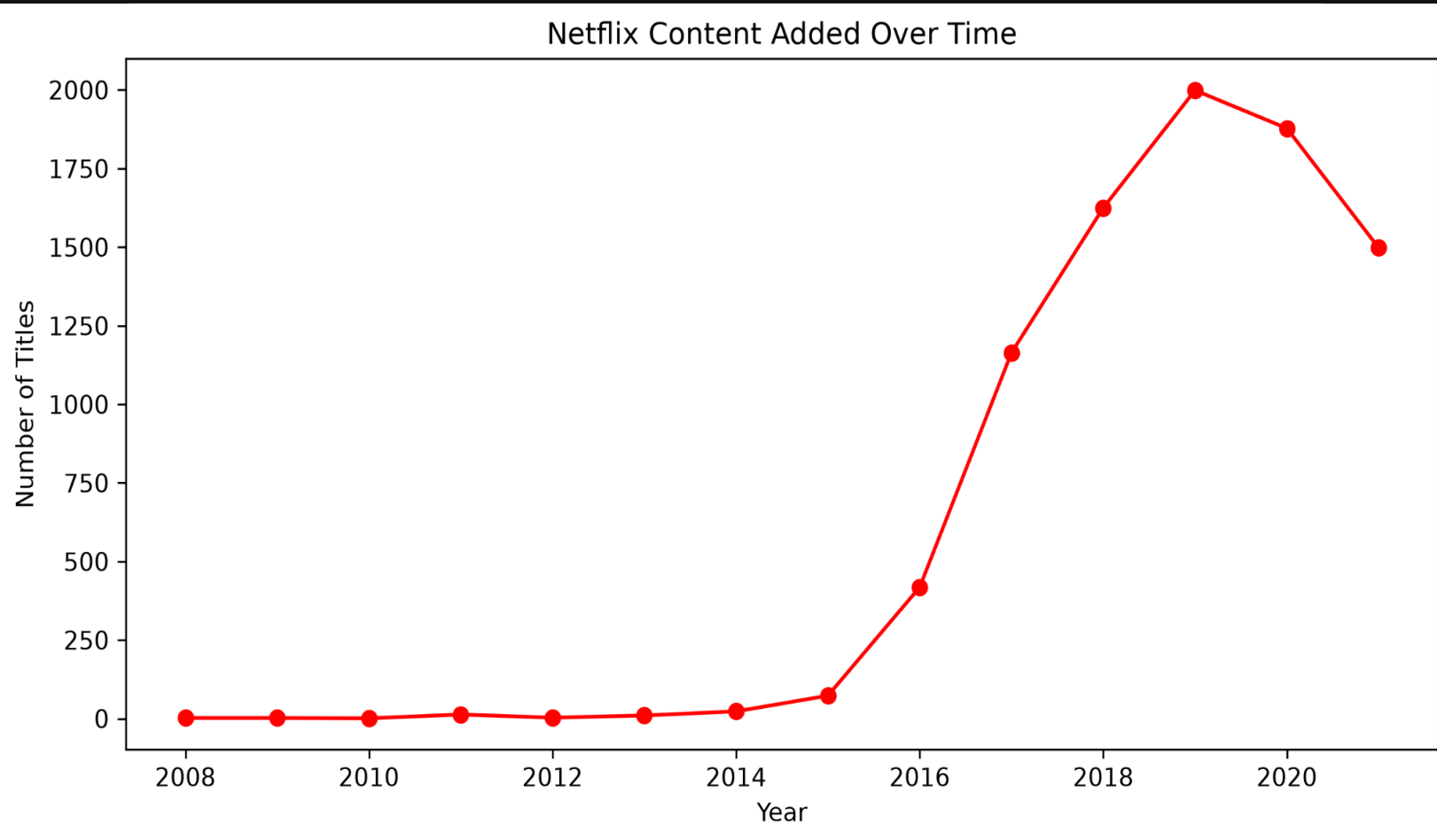
The analysis is guided by questions, not charts

- 1 Which type of content is most common — Movies or TV Shows?
- 2 Which ratings dominate, and does the pattern differ by content type?
- 3 How has the volume of content added changed year over year?
- 4 How does release-year distribution compare with when titles were added?
- 5 Which directors have the most titles?

Finding 01 — Content Added Over Time

08

When did Netflix add the most titles?



2019 ≈ 2,000 titles

KEY PATTERN

Additions stayed under 100 titles/year through 2014, then accelerated sharply from 2015 onward.

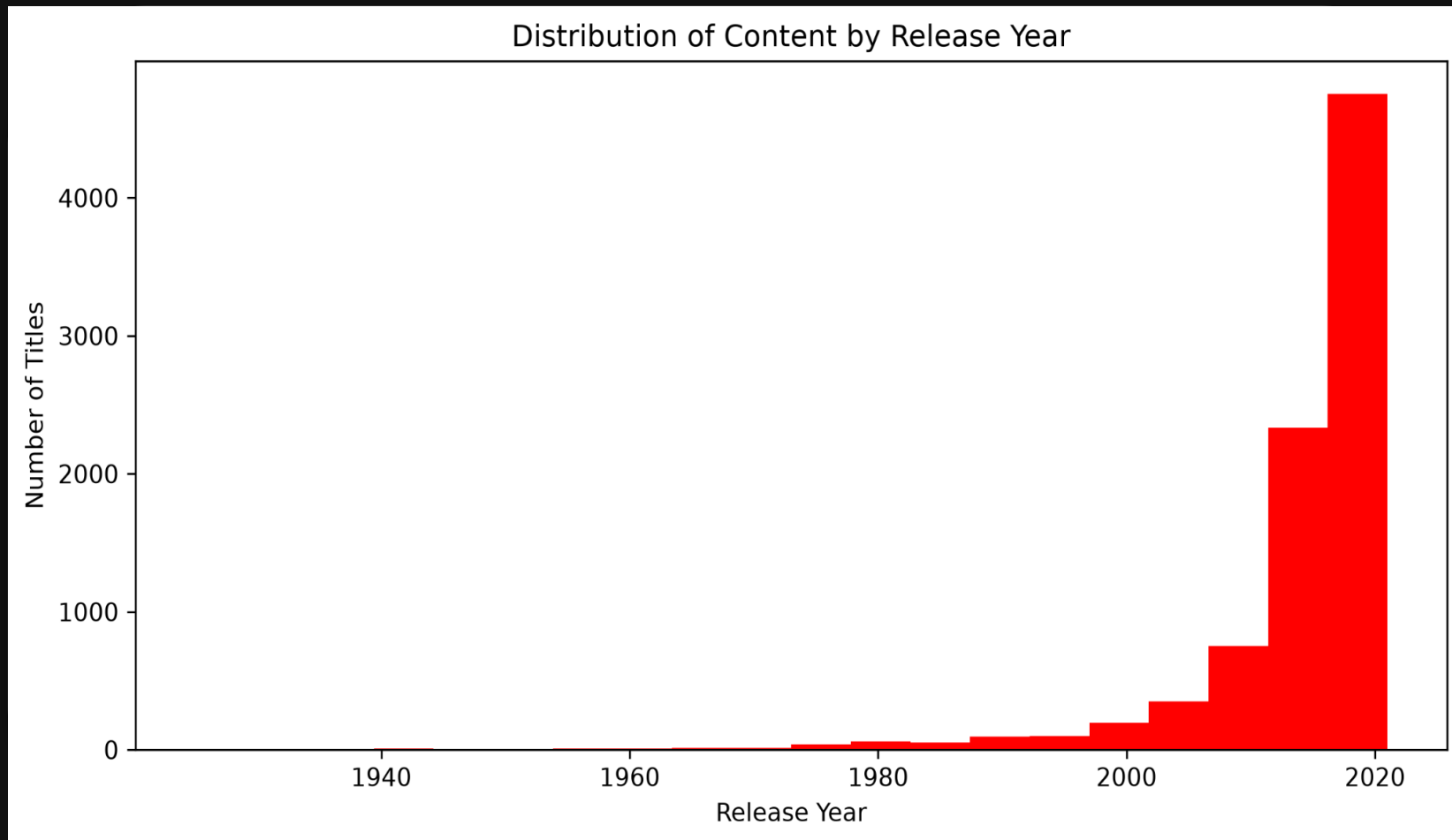
BUSINESS INTERPRETATION

The biggest expansion occurs in 2016–2019, followed by a decline in 2020–2021. The project flags this as a possible strategic pivot rather than simply a slowdown.

Finding 02 — Release Year Distribution

09

How old or recent is the content in the catalog?



**≈ 4,700
recent-release
titles**

KEY PATTERN

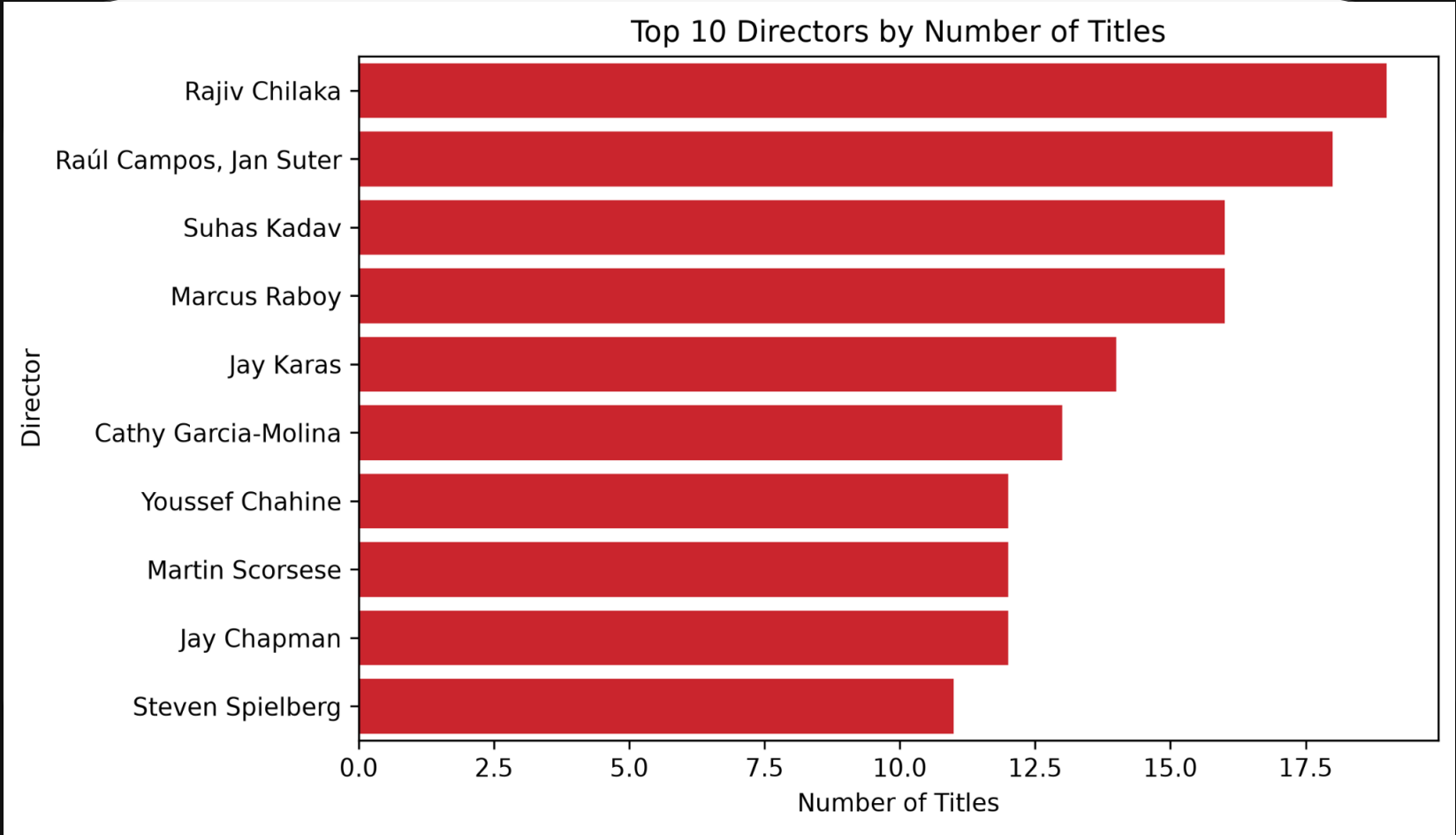
Most titles were originally released after 2010, while very little content predates 1990.

BUSINESS INTERPRETATION

The catalog is strongly modern-content oriented rather than a deep classics library — a useful point when thinking about catalog positioning.

Finding 03 — Top Directors

Which directors appear most frequently in the catalog?



Top: $\approx$ 19 titles

KEY PATTERN

Rajiv Chilaka leads at roughly 19 titles, followed by Raúl Campos & Jan Suter at roughly 18.

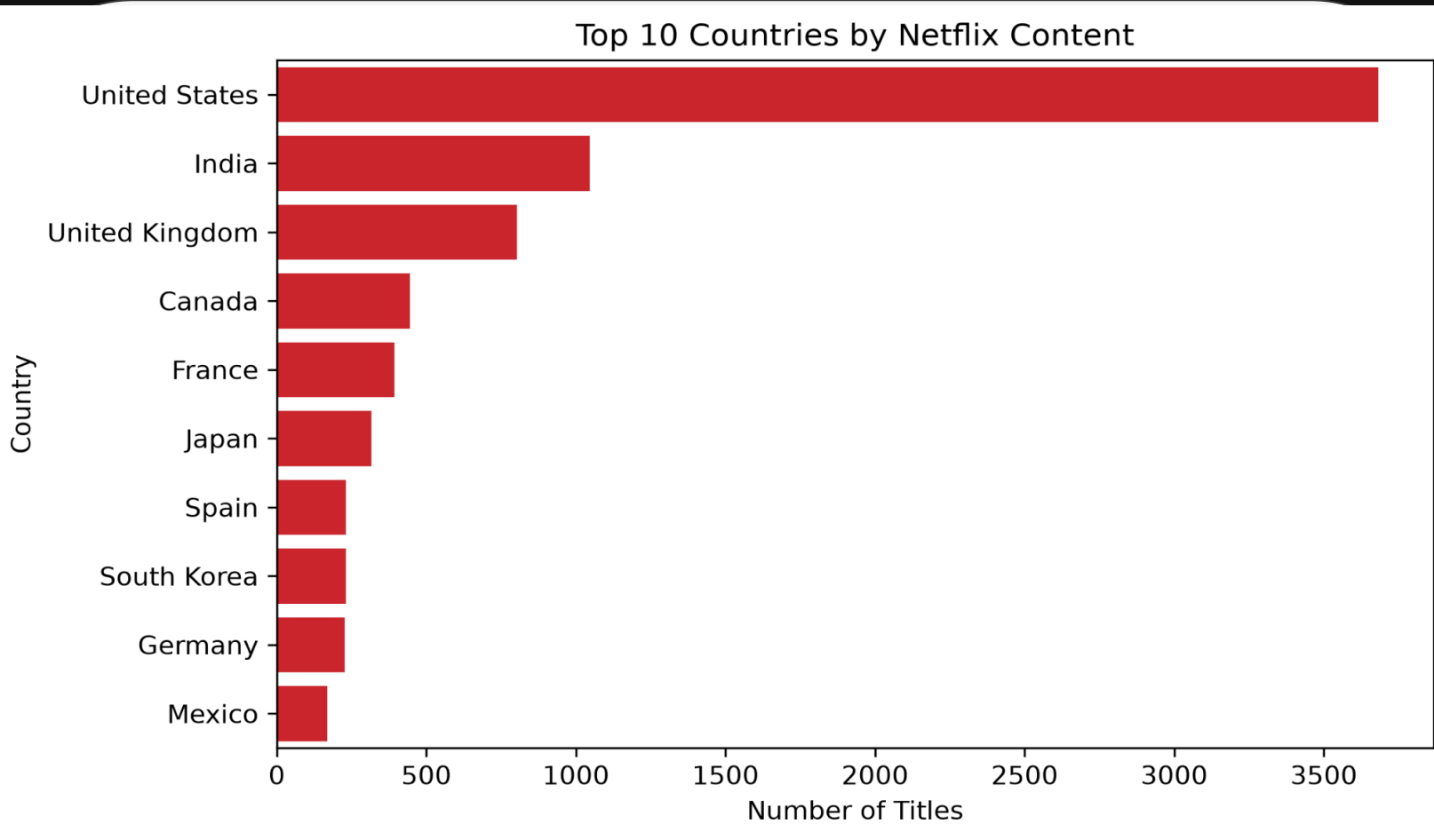
BUSINESS INTERPRETATION

High-output creators can reveal scalable regional production pipelines. The source notes that co-director rows were treated as combined entries.

Finding 04 — Geographic Content Mix

11

Which countries contribute the most titles?



USA #1
India #2

KEY PATTERN

The United States contributes the largest volume, with India second and the United Kingdom third in the chart.

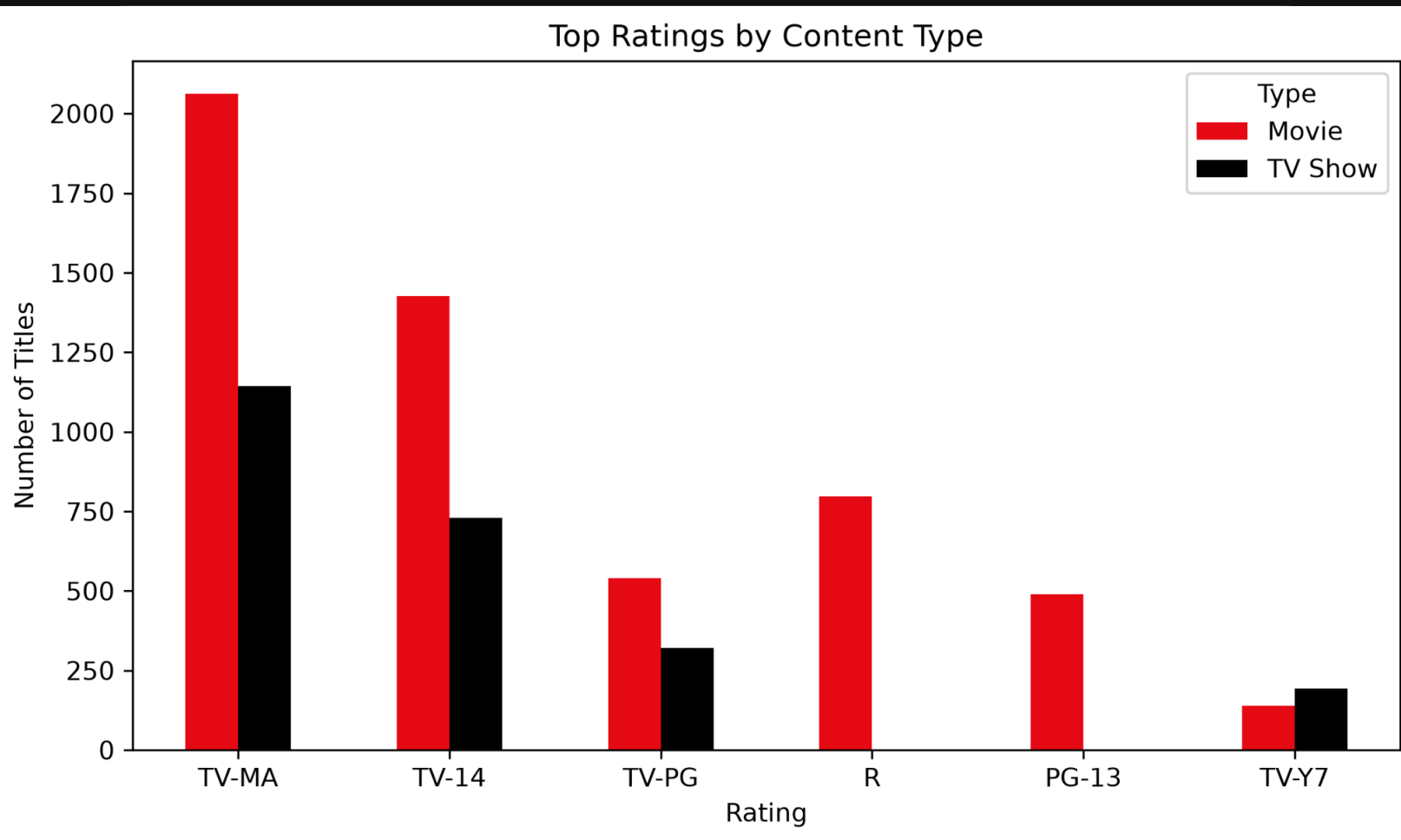
BUSINESS INTERPRETATION

The geographic mix shows why international and regional production pipelines are important to Netflix's catalog strategy.

Finding 05 — Ratings by Content Type

12

How does the rating mix differ between Movies and TV Shows?



TV-MA dominates

KEY PATTERN

TV-MA and TV-14 dominate. In the major rating brackets shown, Movie counts are substantially higher than TV Show counts.

BUSINESS INTERPRETATION

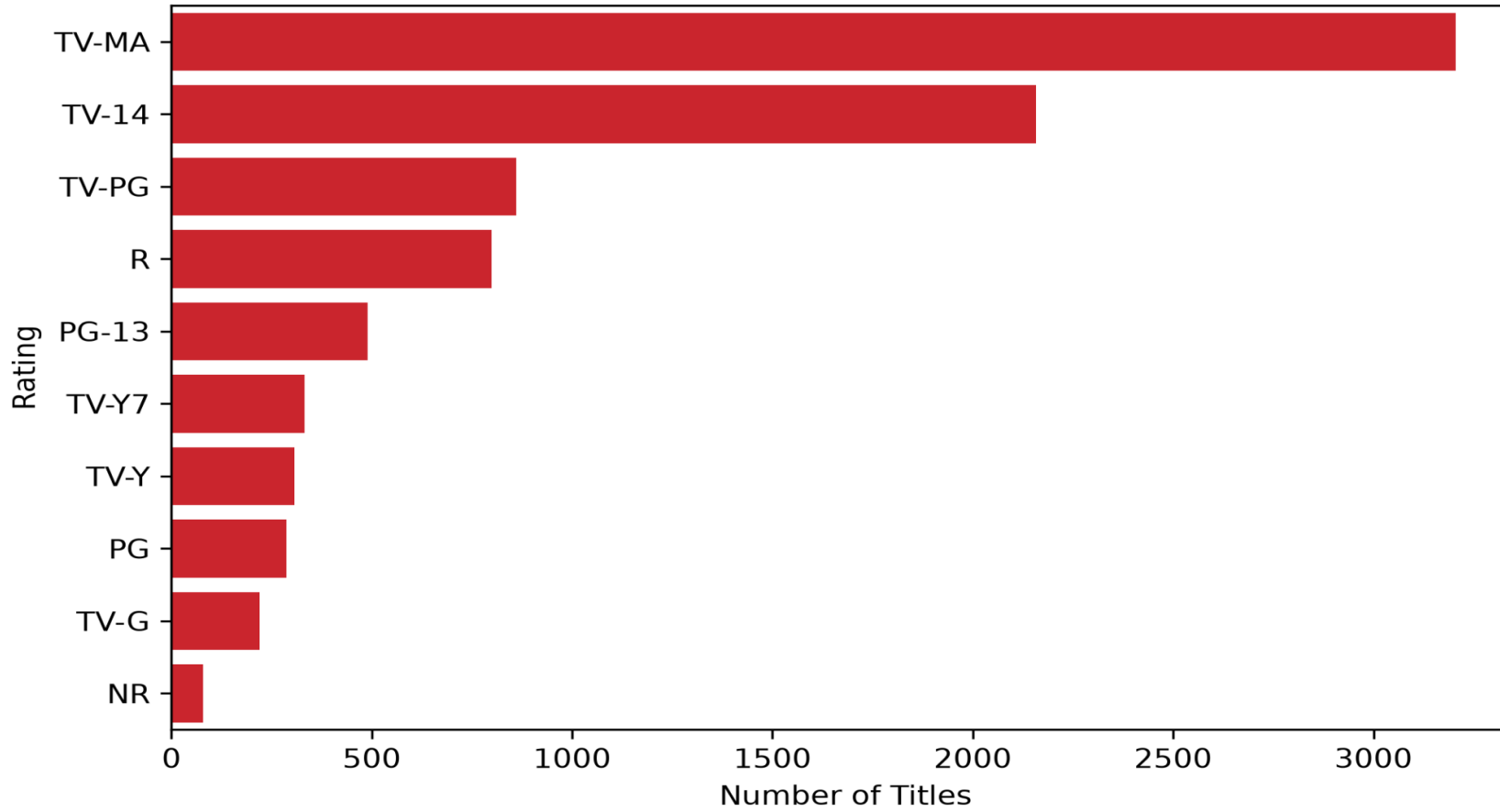
The project interprets the catalog as movie-heavy, while noting that TV Shows held up better than Movies during the post-2019 pullback.

Finding 06 — Overall Content Ratings

13

Which audience ratings dominate the catalog?

Top 10 Content Ratings on Netflix



TV-MA $\approx$ 3,200

KEY PATTERN

TV-MA is the most common rating at roughly 3,200 titles, followed by TV-14 at roughly 2,160.

BUSINESS INTERPRETATION

The catalog skews toward mature audiences. Family-friendly ratings such as TV-Y, TV-G and PG are much smaller.

What the EDA Tells Us

14

Connecting individual findings into one story

CONTENT MIX

Movies significantly outnumber TV Shows.

AUDIENCE

TV-MA and TV-14 are the strongest rating groups.

GROWTH

Catalog additions accelerated sharply after 2014 and peaked around 2019.

RECENCY

Most titles were released after 2010.

GEOGRAPHY

The US leads content volume, followed by India.

CREATORS

A small group of high-output directors stands out in the catalog.

The strongest narrative: Netflix's catalog is large, modern, movie-heavy and concentrated in mature ratings, with clear international and creator-level patterns.

Recommendations

15

Turning EDA observations into strategic discussion

01

Balance the movie-heavy catalog Evaluate opportunities for more series/original investment while maintaining the movie base.

02

Explore the family-content gap Family/kids ratings are much smaller than TV-MA/TV-14, suggesting room to expand if strategically relevant.

03

Scale regional pipelines High-output directors and strong non-US content volumes point to the value of regional production.

04

Track catalog strategy over time The 2019 peak and subsequent decline should be monitored as a possible change in acquisition/original-content strategy.

Technology Stack

17

Tools used to execute the analysis

PYTHON 3

Core analysis environment

PANDAS

Data manipulation and cleaning

NUMPY

Numerical operations

MATPLOTLIB

Static plotting

SEABORN

Statistical visualizations

JUPYTER NOTEBOOK

Interactive EDA workflow

The full analysis and code are maintained in the Netflix EDA.ipynb notebook.